Village Ecological Register

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Section - 1 Introduction:

What is a Village Ecological Register?

The Village Ecological Register is a document prepared by community members, systematically capturing information about a village's traditional ecological knowledge - systems and practices and local biodiversity. It serves as a repository for accumulating knowledge, insights, and data, providing a comprehensive record of socio-ecological changes that happened in the village over time.

This participatory approach will encourage the direct involvement of the community in observing and recording environmental changes. This will also serve as a springboard for stimulating ecological orientation and sensitivity amidst the community members. It will also provide an opportunity for initiating comprehensive discussions within the community concerning ecological indicators.

Why is there a need for preparing a Village Ecological Register?

The primary aim of instituting a village ecological register is to proactively involve communities in recording the oral traditional ecological knowledge systems and practices along with it's rich biodiversity and the ecological changes that have unfolded across the village's history. A pivotal component of this is the preservation of oral village history, serving as an invaluable resource that offers insights into the intricate ecological fabric woven over generations. By incorporating traditional knowledge and anecdotal accounts, the ecological register becomes a means of not only documenting changes but also gaining a profound understanding of how social and ecological changes have happened in the village over time.

Furthermore, the ecological register facilitates the transfer of intergenerational knowledge, ensuring that traditional wisdom and practices are passed down through the ages. This not only enriches the cultural fabric of the community but also contributes to more informed and sustainable ecological practices. Therefore, the initiative would help in fostering citizen initiatives centered around conservation and support in promoting awareness and discussions of commons and biodiversity conservation.

This would play a crucial role in empowering villages to advocate for their rights over natural resources, especially within the framework of Community Forest Rights (CFR). Also, the ecological register supports communities preparing conservation and restoration plans tailored to the village's specific needs. This includes both site-specific and species-specific conservation strategies, ensuring a more targeted and effective approach.

How is it different from the People's Biodiversity Register?

The Village Ecological Register (VER) and People's Biodiversity Register (PBR) diverge fundamentally in their approach and purpose. Village Ecological Register is localized at the village level, employs qualitative narratives, stories, and oral histories, emphasizing traditional ecological knowledge through a participatory approach. It fosters curiosity and ecological sensitivity within the community, promoting a sense of ownership. It can also be potentially used in aiding in Community Forest Rights (CFR) claims. In contrast, PBR, prepared at the Panchayat level, is a technical document generated through a top-down process.

Section - 2 General Information of Village

Village Name: Gram Panchayat: Block:

State: Village (Latitude/Longitude): Date:

2.1 Land details

(Instructions: Gather data for this table through key informant interviews)

Total Village Area (in ha)	Forest Land (in ha)	Grazing Land (in ha)	Revenue Wasteland (in ha)	Agricultural Land (in ha)	Other (in ha)

2.2 Caste composition

(Number of households)

General Category (GEN)	**Scheduled Caste (SC)**	**Scheduled Tribes (ST)**	**Other Backward Castes (OBC)**

Total Population: _____

2.3 Landholding wise details

Category	Large Farmers (More than 10 Ha)	Medium Farmers (4-10 Ha)	Semi Medium (2-4 ha)	Small Farmers (1-2 Ha)	Marginal Farmers (less than 1 Ha)	Landless Farmers
Number of Households						

2.4 Major Livelihoods Sources

(a)

(b)

(c)

(d)

(e)

(f)

(g)

(h)

Section - 3 Village History and local conservation ethos/belief/proverbs

3.1 Documenting Village History:

(Instructions: Prepare a timeline on a chart paper documenting key events, milestones and other developments from past 50-80 years within the village by conducting a Focus Group Discussion.)

3.2 Myths & Popular Beliefs surrounding the village

(Instructions: Ask about the origin of the village, any specific reasons behind the village's name, *are there any traditional methods of documenting and storing knowledge (songs, proverbs, painting, etc. by conducting a focus group discussion), document the name and significance.*

Traditional methods:

Category (song/ painting/ poem)	*Name of Song*	*What does the song signify*	*Who is expert in village/ or know about it in detail*	*Contact Number*

*The contact information can be used to create data base of these artists / also to present their art into local conservation forum.

Popular myths/Belief

Popular myths/ belief	Meaning	Significance/ why it was popular

3.3 - Village Resource Map

(Instructions: In a Chart Paper, map out the village boundary it's key features including it's different land types, land uses and key resources with the help of the community members by conducting a Focus Group Discussion)

Section - 4 Agro-ecological Knowledge and Practices

4.1. Cropping Pattern

(Instructions: Gather data for this table by conducting a Focus Group Discussion (FGD) with a farmer group using chart papers. Subsequently, record the collected information in the table.). Information can be collected using pie charts. Write the name of crop on top for which cultivation area is high and then low and so on...

Seasonal Cropping Pattern	Current	10 years ago	25 years ago	50 years ago	Reason for changes in Cropping patterns
Kharif Season					
Rabi Season					
Zaid Season					

4.2 Traditional crop varieties and their significance

(Instructions: Collect information by conducting a focus group discussion in a tabular format, focusing on the traditional varieties of crops/ seeds, their availability and significance. Label and store these seeds in zip bags, which can later be showcased during the biodiversity fair.)

Traditional Crop Varieties	Availability (Abundant, Scarce, Rare)	Significance (Why are these varieties unique)	Why not prevalent currently (reason)
Example: **Lokwan (variety of wheat)**	Rare	favored for its good chapati-making qualities and good drought tolerance and was well-adapted to the semi-arid regions of western India	Farmers have shifted to modern varieties.

4.3 Farming practices including Traditional practices

Natural farming/ Bullock Farming/ Pest Control/ Seed Preservation and Use for the next cropping season/ seed exchange / shifting cultivation

Instructions: The information in this table is to be collected through focused group discussion with farmers groups on chart paper. The para-ecologist will be provided with relevant resources to carry along with them to the field.

Farming System Practices	Importance	Reasons for diminished prevalence
Example: Planting coriander near susceptible crops	Coriander flowers attract beneficial insects such as hoverflies and wasps, which prey on aphids, caterpillars, and other pests.thus help enhance biological pest control.	More focus on hybrid varieties

4.4 Hedge Biodiversity

Hedge flora and fauna/ birds/ reptiles/ birds /insect/soil Macrofauna

(Instructions: The information in this table is to be collected through focused group discussion with farmers group on chart paper. The Para-ecologist will be provided with pocket guides on birds, dragonflies, butterflies, reptiles, local trees list along with him to initiate the discussion.)

Group	Local Name	Changes in the number over 25 years	Reasons for Changes	Are they Important for Farming? (Yes, No, Don't Know)	Why are they Important?
Insect	*Honey bee*	*Decrease*	*Due to cutting of trees beehives are very less* *Use of fertilizers*		*Crop production is better in the farms where bees visit more frequently. Also, this reduces dependency on hybrid seeds*
Birds	Sparrow	Decrease	Loss of habitat	Yes	*Sparrows are voracious insect predators and feed on a wide variety of insect pests that can damage crops.*

4.5 Soil Health Status:

Soil type, fertility changes (community perception- improved / reduced-reasons for change) / presence and absence of soil macrofauna and other important ecological indicators such as bees/ snakes/ earthworm (now /ten year/ 15 years- reasons of increase or decrease).

(Instructions: Gather data for the following table through focus group discussions)

Soil types	
Soil fertility (status changed over past 20 years - improved/ decreased/ no changed)	
Reasons for changes in soil fertility	

4.6 Soil Health Indicators

What indicator do they look at to understand the soil health status? How have these indicators changed over the past 25 years?

(Instructions: The information of this table is to be collected through focused group discussion with farmers group in table. Para-ecologist will carry Prakriti ka Pitara kit to initiate the discussion in the field.)

Indicators	Significance	Changes from past 25 years	Reason for change
Example: Presence of Earthworm	Soil Nutrient Cycling	Decrease in numbers	Excess use of fertilizer

4.7 Traditional Practices /fairs/ Knowledge related to soil and Climate

List of traditional practices which help people to take informed decisions on crop choices based on moisture level in soil/ rainfall/climate resilience practices/ extreme climatic events/ sustainable harvesting practices:

(Instructions: The information is to be collected through focused group discussion with farmers groups on paper with some examples and later documented in a tabular format.)

List of Traditional Practices	Significance
Bidri Festival	Just before monsoon, the festival is organized in Mandla. Different varieties of seeds are collected, offered to the local deity. In the festivities, seeds are exchanged…

4.8 Agriculture Pest Incidences

(Instructions: The information is to be collected through focused group discussion with farmers group on paper with some examples and later documented in a tabular format.)

Name of the Crop	Name of the Disease	Major Pests	Season	Frequency (Seasonal, once in 5 Years, once in 10 Years)	What does a farmer do?	Natural pest controller or traditional practices
Maize	Kali Latt	Kali Latt	Kharif	Sesonal	Use of Pesticide	NA

4.9 Major weeds in agriculture and their control methods

(Instructions: The information is to be collected through focused group discussion with farmers groups on paper with some examples and later documented in a tabular format).

Name of the weeds	Name of the crop	Control methods used by the farmers	Any other traditional methods
Wild oat	Wheat	Hand pick, crop competition	***Livestock Grazing:*** *Livestock, such as sheep or cattle, can selectively target weeds while providing additional income or resources for farmers*

Section - 5 Livestock

5.1 Livestock number

(Instructions: The information is to be collected using focused group discussion with livestock rearers/ farmers and is later documented in a tabular format). Insert number for current year and for 10 years, 20 year and 50 years ago insert (↑***- Increase*** ↓**decrease** ↔ **remain same)**

Livestock	Current Numbers in village	10 years ago ↑ ↓ ↔	25 Years Ago ↑ ↓ ↔	50 years Ago ↑ ↓ ↔	Reasons for Change (?)
Cows (indigenous)					
Cows (Hybrids)					
Oxen					
Camel	Number (x)				
Buffaloes					
Goats					
Sheeps					
Others					

5.2 Type of Indigenous breeds and changes in their number over time

Insert number for current year and for 10 years, 20 year and 50 years ago insert (↑- Increase ↓decrease ↔ remain same)

Indigenous Breed Name (Type)	Current (Numbers)	10 years ago ↑ ↓ ↔	25 Years Ago ↑ ↓ ↔	50 years Ago ↑ ↓ ↔	Reasons for Change (?)
Jamnapari (Goat)					
Sirohi (Goat)					
					`

5.3 Common Livestock Diseases

(Rare < Uncommon < Common)

Name of the Diseases	Livestock Category	Season of outbreak	Current Status (category)	10 years ago ↑ ↓ ↔	25 Years Ago ↑ ↓ ↔	50 years Ago ↑ ↓ ↔	Reasons for Change (?)
Foot and Mouth Disease	Cow	Monsoon	Common	↑	↑	↓	Disease outbreak

5.4 Traditional Ethno-veterinary Practices:

Mention the name of livestock diseases and local practice use to cure disease

Name of the Diseases	Name of the Medicinal Plant/Animals/ Fungi that is use to cure the disease	Name of the Parts
Haadjod		

5.5 Traditional Knowledge & Customary Practices Surrounding Livestock Keeping

Practice	Relevance
Jallaikattu	Conservation of Indigenous breed

Section - 6 Waterscape

*(Instructions: Gather data for the following tables through focus group discussion with mixed group). Example if there are a total of 10 working open well in the village and on an average water is available entire 12 months for drinking. Then **tick** √ in 12 months column and write 10 in total number column*

6.1 Availability of Water for Drinking & Domestic Usage

Source	Total Numbers	12 Months (1)	11-8 Months (2)	7- 4 Months (3)	< 4 Months (4)
Tap					
Tube-well					
Open Well					
Spring					
River					
Other					

6.2 Availability of Water for Livestock purposes

Source	Total Numbers	12 Months (1)	11-8 Months (2)	7- 4 Months (3)	< 4 Months (4)
Tap					
Tube-well					
Open Well					
Spring					
River					
Other (Specify)					

6.3 Source of water for Irrigation

Major Water Sources for irrigation: _____

_____

6.4 Changes into water availability

Has Water Quality (e.g. salinity/hardness) changed over time? (Yes improved/ yes deteriorate, no change)

Source	Changes yes mproved ↑ yes leteriorate ↓, no change) ↔	Before 10 year ↑ ↓, ↔	Before 25 year ↑ ↓, ↔	Before 50 year ↑ ↓, ↔	Reason for Change
Surface water (Pond/River/other)					
Ground water (Tubewell/Open well/Spring					

6.5 Traditional Methods of water conservation

List down and detail the traditional methods of water conservation if any exist in the village.(Tanka, Johad, Pokhar)

6.6 Detail of Important water bodies/wetlands which is ecological important for village

Please mention name, location (image with Geo codes) and why the water body is unique and important.

Example: Sushila sagar talab; Home of lots of migratory birds

Section – 7 Forest Lands

(Instructions: Gather data for this section through focus group discussion with a mixed group), Take a good quality image of forest land using notecam for geocodes). The forest form is to be filled for each of the forest patch. (need to be filled in tabular form)

7.1 General information

- Name of the forest:
- Type of Forest (RF/ PF/ Revenue/ Others):
- Location: (Major landmark/ Geo codes):
- Size of the Forest (in ha):
- Map of the forest (made by community members) - Fire/Invasive/NTFP/Biodiversity/Density

7.2 Major Plant Species Composition of forest:

Top four or five species in each of the category. This table is to be filled for each of the forest patch. (Write the name of species as per the ranking / or abundance in the forest). The table is to be filled for each patch of the forest.

Name of forest_____

Groups	Current Status	10 years ago	25 Years Ago	50 years Ago	Reasons for Change?
Tree					
Shrubs					
Herbs					
Grasses					
Climbers					

7.3 NTFP availability in the forest

Name of forest_____

(Now/10 years/25 years/ 50 years). The table is to be filled for each patch of the forest.

NTFP (fuel-wood fodder and wild edibles or others)	Current availability (very good, medium very less)	10 years ago ↑ ↓, ↔	25 Years Ago ↑ ↓, ↔	50 years Ago ↑ ↓, ↔	Reasons for Change?

Section - 8 Grassland/ Grazing land

This information is to be filed for each patch of grazing land.

Name of grassland / grazing land_____

8.1 General information

- Name of the Grazing land:
- Location: (Major landmark):
- Size of the grazing land (in ha):
- Map of the Grassland (made by community members)
  Fire/Invasive/NTFP/Biodiversity/Density

8.2 Major Plant Species Composition of grassland/ gazing land

Name of the grazing land/ grassland_____

Groups	Current Status	10 years ago ↑ ↓, ↔	25 Years Ago ↑ ↓, ↔	50 years Ago ↑ ↓, ↔	Reasons for Change?
Tree					
Shrubs					
Herbs					
Grasses					
Climbers					

8.3 Major NTFP collected from the Grassland/ grazing land in the village

Availability (Now/10 years/25 years/ 50 years)

Name of the grazing land/ grassland_____

NTFP (fuelwood fodder and wild edibles or others)	Current availability (very good, medium very less)	10 years ago ↑ ↓, ↔	25 Years Ago ↑ ↓, ↔	50 years Ago ↑ ↓, ↔	Reasons for Change?

Section - 9 Revenue Waste Land (RWL)

- The information of revenue waste land is to be filled for each patch of the revenue waste land.
- Name of the patch of revenue wasteland_____

9.1 General information

- Name of the RWL:
- Location: (Major landmark):
- Size of the RWL (in ha):
- Map of the RWL (made by community members) Fire/Invasive/NTFP/Biodiversity/Density

9.2 Major Plant Species Composition of RWL

Groups	Current Status	10 years ago	25 Years Ago	50 years Ago	Reasons for Change?
Tree					
Shrubs					
Herbs					
Grasses					
Climbers					

Section - 10 Grooves/ Banni/ Sacred groove

(Instructions: Gather data for this section along with name of the sacred groove, location including image and any local knowledge and stories about its cultural significance through focus group discussion with a mixed group)

Section – 11 Ecologically important sites

(*Instructions: gather data on ecologically important sites such as waterbody/ wetland with high birds / dragonfly diversity, A trees where lots of egret/ parakeets' roost, Hornbills on trees, etc. through group discussion) : Take photo of that site using notecam for geocodes.*

Section – 12 List of Old and Giant trees in the village

Location and name

(Instructions: Gather data for this section along with name and location of the trees through focus group discussion with a mixed group). Take a image using notecam for geocode.

Name of Tree	Geocodes	Landmark (example hanuman Mandir ke paas)/ remarks

Section – 13 Locations of big bee hives in the villages

Location

(Instructions: Gather data for this section along with name and location of the trees through focus group discussion with a mixed group): Take image using notecam for geocode.

Location	Geo codes	Landmark/Remarks

Section – 14 Fire incidence in village

(Instructions: Gather data for this section through focus group discussion with a mixed group example-When it get happened, frequency, damage/ loss, role of community , Forest department etc)

Section – 15 Local Conservation ethos

(Instructions: Gather data for this section through focus group discussion with a mixed group)

Local conservation ethos refers to the set of values, beliefs, and practices that guide communities in their efforts to conserve and sustainably manage natural resources and ecosystems at the local level. It encompasses the cultural, social, economic, and environmental considerations that influence how communities interact with their natural surroundings and strive to protect biodiversity and ecological integrity.

Section - 16 Medicinal plants, their uses and availability

(Instructions: Gather data for this section through focus group discussion with a mixed group)

Name of Plant	Uses	Changes in availability over past 25 years ↑, ↓, ↔

Section - 17 Invasive plants in the village

(Instructions: Gather data for this section through focus group discussion with a mixed group)

Name of Plant	When It started appearing in village	Harmful impact on communities	Changes in availability over past 25 years ↑ ↓, ↔
Example: Hyptis suaveolens (van tulsi)	15 years	Fodder reduction	Increasing

Section - 18 Feral animal in the village

(Instructions: Gather data for this section through focus group discussion with a mixed group)

Name of Animal	Harmful impact on communities	Changes in availability over past 25 years
Example: Feral cattle	Crop loss	Increasing

Section - 19 socially/ culturally protected plants and animals

(Instructions: Gather data for this section along with their name and why they are protected through focus group discussion with a mixed group)

Name	Category	Cultural Significance
Bargad	Tree	Religious purposes/ worship

Section - 20 List of flora and Fauna

20.1 Tree diversity

(Tick in the column number 4 to 8 if the species present in the mentioned landuse). Prepare the herbarium of species which you are not able to identify or not exist in the reference material.

Local Name (1)	Flower color (2)	Use/ traditional (3)	Forest (4)	Grazing land (5)	Revenue Wasteland (6)	Near Water bodies (7)	In and around agriculture (8)	Changes in availability over past 25 years ↑ ↓, ↔ (9)
Palash	Orange	make color during the festival of Holi	√				√	↓

20.2 Shrub diversity

(Tick in the column number 4 to 8 if the species present in the mentioned landuse). Prepare the herbarium of species which you are not able to identify or not exist in the reference material.

Local Name (1)	Flower color (2)	Use/ traditional (3)	Forest (4)	Grazing land (5)	Revenue Wasteland (6)	Near Water bodies (7)	In and around agriculture (8)	Changes in availability over past 25 years ↑ ↓, ↔ (9)

20.3 Herbs and Grass diversity

(Tick in the column number 4 to 8 if the species present in the mentioned landuse). Prepare the herbarium of species which you are not able to identify or not exist in the reference material.

Local Name (1)	Flower color (2)	Use/ traditional (3)	Forest (4)	Grazing land (5)	Revenue Wasteland (6)	Near Water bodies (7)	In and around agriculture (8)	Changes in availability over past 25 years ↑ ↓, ↔

20.4 Lower plants bryophytes

(mosses and liverworts), lichens, fungi and algae (including diatoms).

Local Name (1)	Herbarium number (2)	Use/ traditional (3)	Forest (4)	Grazing land (5)	Revenue Wasteland (6)	Near Water bodies (7)	In and around agriculture (8)	Changes in availability over past 25 years ↑ ↓, ↔ (9)

20.5 Mammal Diversity

Local Name	Name in guide	Habitat	Name of the place in village where it is mostly seen	Changes in availability over past 25 years ↑ ↓, ↔	Reasons for changes	Remarks (associated TEK)
Tiger	Bagh	Forest	Village forest	↓	Loss of habitat	Traditionally workshiped

20.6 Bird Diversity

Local Name	Name in pocket guide	Habitat	Name of the place in village where it is mostly seen	Changes in availability over past 25 years ↑ ↓, ↔	Reasons for changes	Remarks (associated TEK)

20.7 Reptiles and amphibians (Snakes, lizard, turtles, frog toads) Diversity

Local Name	Group	Habitat	Name of the place in village where it is mostly seen	Changes in availability over past 25 years ↑ ↓, ↔	Reasons for changes	Remarks (associated TEK)

20.8 Butterflies Diversity

Local Name	Name in photo guide	Habitat	Name of the place in village where it is mostly seen	Changes in availability over past 25 years ↑ ↓, ↔	Reasons for changes	Remarks (associated TEK)

20.9 Dragonflies Diversity

Local Name	Name in photo guide	Habitat	Name of the place in village where it is mostly seen	Changes in availability over past 25 years ↑ ↓, ↔	Reasons for changes	Remarks (associated TEK)

20.10 Fishes/Insect/ spider/or any other groups

Local Name	Group	Habitat	Name of the place in village where it is mostly seen	Changes in availability over past 25 years ↑ ↓, ↔	Reasons for changes	Remarks (associated TEK)

20.11 Soil Macrofauna

Local Name	Group	Habitat	Name of the place in village where it is mostly seen	Changes in availability over past 25 years ↑ ↓, ↔	Reasons for changes	Remarks associated TEK)